

Customer Behavior Analysis Using Machine Learning

Project Title

Customer Behavior Analysis and Customer Segmentation for Alfido Tech

1. Introduction

Customer behavior analysis is an important process in modern businesses for understanding customer purchasing habits, identifying customer segments, predicting churn behavior, and improving customer engagement strategies. Businesses collect large amounts of transactional data, but meaningful insights can only be extracted through proper data analysis and machine learning techniques.

This project focuses on analyzing customer transaction and behavioral data using Python, data visualization, and clustering algorithms. The project aims to identify different customer groups, understand purchasing patterns, analyze customer churn, and provide business recommendations for improving customer retention and revenue generation.

The analysis was performed using exploratory data analysis (EDA), feature engineering, RFM analysis, and K-Means clustering.

2. Objectives

The major objectives of this project are:

- To clean and preprocess customer transaction data
 - To analyze customer purchasing behavior
 - To identify customer segments using clustering techniques
 - To study customer churn patterns
 - To visualize customer retention and sales trends
 - To provide actionable business recommendations for Alfido Tech
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3. Dataset Description

The dataset used in this project contains customer transaction and demographic information.

Dataset Columns

Column Name	Description
Customer ID	Unique ID of customers
Purchase Date	Date of transaction
Product Category	Category of purchased product
Product Price	Price of product
Quantity	Quantity purchased
Total Purchase Amount	Total amount spent

Column Name	Description
Payment Method	Method of payment
Customer Age	Age of customer
Returns	Product return information
Customer Name	Name of customer
Age	Age information
Gender	Gender of customer
Churn	Customer churn status

The dataset provides valuable information related to customer demographics, purchasing behavior, and transaction history.

4. Technologies Used

The following technologies and libraries were used in this project:

Programming Language

- Python

Libraries Used

- Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn
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5. Data Preprocessing

Data preprocessing is one of the most important steps in data analysis because raw datasets often contain missing values, duplicate entries, and inconsistent formats.

The following preprocessing steps were performed:

5.1 Handling Missing Values

Missing values were filled using forward fill techniques to maintain data consistency.

5.2 Removing Duplicate Records

Duplicate rows were removed to improve dataset quality.

5.3 Column Renaming

Spaces in column names were replaced with underscores for easier coding and manipulation.

5.4 Date Conversion

The Purchase Date column was converted into datetime format for time-based analysis.

5.5 Feature Engineering

Additional features were created such as:

- Month of purchase
- Day name
- Total spending

These features helped improve the analysis process.

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand customer purchasing behavior and identify patterns within the dataset.

6.1 Customer Spending Distribution

A histogram was used to analyze the distribution of customer spending.

Observation

- Most customers spend moderate amounts.
 - A small number of customers spend significantly higher amounts.
 - The spending pattern is positively skewed.
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6.2 Monthly Sales Trend

Monthly sales trends were analyzed using line charts.

Observation

- Sales fluctuate across different months.
 - Some months show higher customer purchasing activity.
 - Seasonal trends may influence customer purchases.
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6.3 Product Category Analysis

Bar charts were used to identify top-performing product categories.

Observation

- Certain categories contribute significantly to overall revenue.
 - High-performing categories can be targeted for marketing campaigns.
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6.4 Payment Method Analysis

Customer payment preferences were analyzed.

Observation

- Customers prefer specific payment methods more frequently.
 - Businesses can optimize payment systems based on customer preferences.
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6.5 Gender Distribution

Gender-based customer analysis was performed.

Observation

- The dataset contains customers from different genders.
 - Gender-based marketing strategies can be developed.
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6.6 Churn Analysis

Customer churn distribution was visualized.

Observation

- Some customers are likely to discontinue purchases.
 - Churn prediction is important for customer retention.
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7. RFM Analysis

RFM analysis is a customer segmentation technique based on:

- Recency
- Frequency
- Monetary Value

7.1 Recency

Measures how recently a customer made a purchase.

7.2 Frequency

Measures how often customers purchase products.

7.3 Monetary Value

Measures the total spending of customers.

RFM analysis helps identify:

- Loyal customers
 - High-value customers
 - At-risk customers
 - Low-engagement customers
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8. Customer Segmentation Using K-Means Clustering

K-Means clustering was used to group customers based on purchasing behavior.

8.1 Data Scaling

Features were standardized using StandardScaler.

8.2 Elbow Method

The Elbow Method was used to determine the optimal number of clusters.

8.3 Cluster Formation

Customers were divided into four segments based on:

- Recency
 - Frequency
 - Monetary value
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9. Segment Analysis

The customer segments identified are:

Cluster 0 – High Value Customers

- Frequent purchases
- High spending
- Loyal customers

Cluster 1 – Moderate Customers

- Average spending behavior
- Moderate purchase frequency

Cluster 2 – Low Engagement Customers

- Less frequent purchases
- Lower spending patterns

Cluster 3 – Churn Risk Customers

- High recency
- Low purchase frequency
- Low monetary contribution

These customer groups help businesses create targeted strategies.

10. Churn Risk Analysis

Churn-risk customers were identified using:

- Low frequency
- Low spending
- High recency

Observation

- Some customers show signs of disengagement.
 - These customers require retention strategies.
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11. Customer Retention Analysis

Retention trends were analyzed using unique customer counts over time.

Observation

- Customer retention varies monthly.
 - Strong retention strategies are required to maintain long-term engagement.
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12. Correlation Analysis

A heatmap was used to study relationships between numerical features.

Observation

- Product price and total spending show positive correlation.
 - Quantity purchased directly impacts revenue.
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13. Business Insights

The project generated several valuable insights:

1. High-value customers contribute major revenue.
 2. Some product categories outperform others significantly.
 3. Churn-risk customers can be identified early.
 4. Customer spending behavior varies across segments.
 5. Returns may affect overall profitability.
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14. Recommendations for Alfido Tech

Based on the analysis, the following recommendations are provided:

1. Loyalty Programs

Introduce loyalty rewards and membership benefits for high-value customers.

2. Personalized Recommendations

Use customer purchase history for personalized product suggestions.

3. Churn Prevention Strategies

Offer discounts and promotional campaigns to churn-risk customers.

4. Focus on Top Categories

Increase inventory and marketing efforts for best-performing categories.

5. Improve Customer Experience

Reduce product returns through better product quality and customer support.

15. Conclusion

This project successfully analyzed customer purchasing behavior using machine learning and data analysis techniques.

Through Exploratory Data Analysis, RFM analysis, and K-Means clustering, important customer segments and purchasing patterns were identified. The project also highlighted churn-risk customers and provided actionable business recommendations for Alfido Tech.

The findings from this analysis can help businesses improve:

- Customer retention
- Personalized marketing
- Revenue generation
- Customer engagement strategies

Overall, this project demonstrates how data analytics and machine learning can support better business decision-making and customer relationship management.

16. Future Scope

Future improvements can include:

- Real-time customer behavior tracking
 - Predictive churn modeling using classification algorithms
 - Deep learning-based recommendation systems
 - Customer lifetime value prediction
 - Integration with business dashboards
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17. References

- Python Documentation
 - Scikit-learn Documentation
 - Pandas Documentation
 - Seaborn Documentation
 - Kaggle Dataset Repository
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18. Dataset Reference

<https://www.kaggle.com/datasets/bhanupratapbiswas/customer-behavior-analysis>